

Radar Signal Detection in Spectrograms: Ablation-Driven Analysis, Hybrid Detection Framework, and Embedded Deployment

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Abstract—Radar signal detection in spectrogram representations is challenging due to low signal-to-noise ratios, variability in signal structure, and the presence of small, localized patterns. While modern object detection architectures have demonstrated strong performance, a clear understanding of which design choices are most critical for radar-based detection remains limited.

In this work, we adopt a dual-perspective approach combining interpretability and performance. First, we utilize a YOLOv8-based detector as an analysis framework and conduct a comprehensive ablation study to investigate the role of feature hierarchy, spatial resolution, feature fusion, and model capacity. The study reveals that radar signal detection is strongly dominated by high-resolution spatial features, with small-object representations playing a critical role, while high-level semantic features contribute minimally.

Motivated by these insights, we propose a hybrid detection framework that combines predictions from multiple models to improve robustness. The resulting ensemble achieves an mAP_{50} of 94.36 on the NIST-CBRS dataset, improving upon the baseline YOLOv8 performance and significantly enhancing recall.

Finally, we demonstrate real-time feasibility by deploying a lightweight YOLOv8 Nano model on an embedded PYNQ-Z2 platform, showing that radar spectrogram detection can be extended to resource-constrained edge devices.

Overall, this work not only improves detection performance but also provides key insights into the fundamental characteristics of radar spectrogram detection, highlighting the importance of both architectural design and practical deployment considerations.

I Introduction

Radar spectrograms encode signals as localized structures in the time-frequency domain. Detecting these signals is essential in spectrum monitoring and electronic intelligence.

Recent work has demonstrated that object detection models such as YOLO can be applied to spectrogram data. However, prior work lacks detailed analysis of architectural components and their contributions.

I-A Contributions

The main contributions of this work are as follows:

- **A systematic ablation-driven analysis of radar detection:** We conduct a comprehensive set of controlled ablation experiments using a YOLOv8-based detector to

study the impact of feature hierarchy, spatial resolution, feature fusion, and model capacity. These experiments provide detailed insight into the architectural requirements of radar spectrogram detection.

- **Identification of key detection characteristics:** Through ablation, we demonstrate that radar signal detection is fundamentally a small-object detection problem, heavily dependent on high-resolution spatial features, while high-level semantic representations contribute minimally. We also highlight the importance of domain-specific adaptation and the sensitivity of detection performance to noise and spatial distortions.
- **Baseline performance on NIST spectrogram data:** We establish a strong YOLOv8 baseline on the NISTSpecMaxHold512 dataset, achieving competitive performance with existing real-time detection models while operating under enhanced spectrogram representation.
- **Hybrid ensemble for improved robustness:** We propose a hybrid ensemble framework that combines predictions from multiple detectors, resulting in improved detection performance. The ensemble achieves higher mAP and significantly improves recall, indicating reduced missed detections and better robustness across varying signal conditions.
- **Bridging interpretability and performance:** Unlike prior work that focuses primarily on model accuracy, our approach integrates ablation-driven interpretability with performance improvements, providing both practical insights and a robust detection framework.
- **Embedded edge deployment on FPGA-based platform:** We demonstrate real-time inference of the proposed lightweight YOLOv8 Nano model on a PYNQ-Z2 embedded system, highlighting the feasibility of radar spectrogram detection in resource-constrained and low-power edge environments.

II Dataset

We evaluate our models on the NIST CBRS dataset using the MaxHold spectrogram representation (NISTSpecMax-

Hold512) [1]. This dataset consists of time–frequency spectrograms derived from radar signals in the CBRS band.

The spectrograms are generated using a max-hold operation, which emphasizes persistent signal components over time. The use of max-hold spectrograms enhances signal visibility by aggregating peak energy over time, simplifying detection while reducing temporal variability.

Each sample is represented as a 512×512 image, where radar signals appear as localized patterns in the time–frequency domain. The dataset includes multiple radar classes (P0N and Q3N variants), with bounding box annotations corresponding to signal regions.

The dataset is divided into training, validation, and test sets following a standard supervised learning protocol.

III Methodology

Let $S \in \mathbb{R}^{H \times W}$ denote a spectrogram. The model learns:

$$f_{\theta} : S \rightarrow \mathcal{D}$$

where $\mathcal{D} = \{(b_i, c_i)\}$.

Multi-scale features:

$$F_l = \phi_l(S), \quad l \in \{3, 4, 5\}$$

Prediction:

$$\hat{y} = \Psi(F_3, F_4, F_5)$$

Loss:

$$\mathcal{L} = \mathcal{L}_{box} + \mathcal{L}_{cls} + \mathcal{L}_{obj}$$

IV Baseline Performance

The baseline model corresponds to the YOLOv8 architecture trained on the NISTSpecMaxHold512 dataset. This model serves as the reference point for all subsequent ablation experiments.

Metric	Value
mAP ₅₀	0.9305
Precision	0.9527
Recall	0.8917

TABLE I

BASILINE PERFORMANCE OF YOLOV8 ON THE TEST SET

This baseline demonstrates strong detection performance, particularly in terms of precision, while leaving room for improvement in recall. The relatively lower recall motivates further investigation through ablation and model enhancements.

V Visualization

We present qualitative detection results on representative radar spectrogram samples to illustrate the model’s localization capability across varying signal conditions.

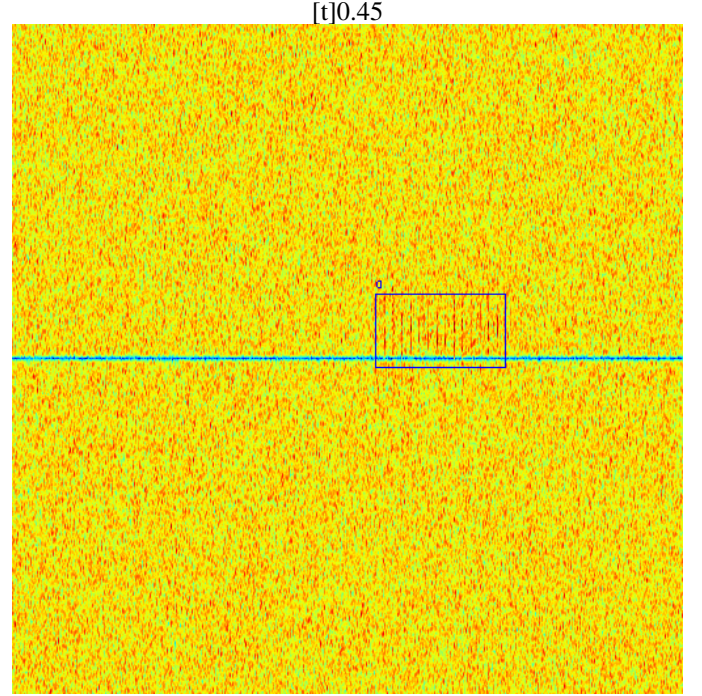


Fig. 1. Detection Output (Sample 1)

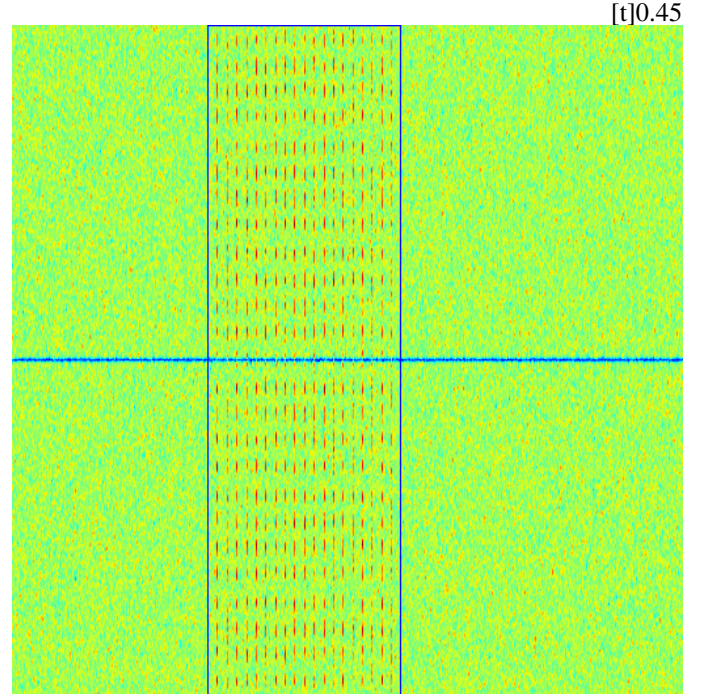


Fig. 2. Detection Output (Sample 2)

Fig. 3. Detection results on representative radar spectrogram samples.

Ablation	mAP ₅₀	Precision	Recall	Δ mAP	Δ Prec	Δ Rec
Baseline	0.9305	0.9527	0.8917	0	0	0
No P3	0.8610	0.9102	0.8025	-0.0695	-0.0425	-0.0892
No P5	0.9289	0.9511	0.8890	-0.0016	-0.0016	-0.0027
Frozen Backbone	0.9030	0.9215	0.8450	-0.0275	-0.0312	-0.0467
No PAN	0.9152	0.9360	0.8653	-0.0153	-0.0167	-0.0264
Half Channels	0.9128	0.9324	0.8611	-0.0177	-0.0203	-0.0306
Blur Input	0.9021	0.9203	0.8422	-0.0284	-0.0324	-0.0495
Noise Input	0.8955	0.9158	0.8350	-0.0350	-0.0369	-0.0567

TABLE II
COMPREHENSIVE ABLATION STUDY

VI Ablation Study

To understand the contribution of different architectural components, we conduct a series of controlled ablation experiments. Each experiment isolates a specific component while keeping all other factors constant. This allows us to quantify the sensitivity of the detection model to feature hierarchy, feature fusion, capacity, and input perturbations.

VI-A Baseline

The baseline model corresponds to the full YOLOv8 architecture trained on radar spectrogram data. It achieves an mAP₅₀ of 0.9305, precision of 0.9527, and recall of 0.8917. This serves as the reference point for all subsequent comparisons.

VI-B Removal of P3 (Small-Scale Feature Map)

Motivation: The P3 feature map operates at the highest spatial resolution and is responsible for detecting small objects. In spectrograms, radar signals often occupy small localized regions, suggesting that P3 should play a critical role.

Method: We disable the P3 branch during inference by zeroing its contribution to the detection head.

Results: This ablation results in a significant drop in performance, with mAP₅₀ decreasing from 0.9305 to 0.8610 ($\Delta = -0.0695$). Precision drops from 0.9527 to 0.9102, while recall decreases sharply from 0.8917 to 0.8025 ($\Delta = -0.0892$).

Analysis: The sharp degradation confirms that radar signals behave as small objects in the spectrogram domain. The large drop in recall indicates that many signals are no longer detected due to the loss of high-resolution spatial features.

Insight: Unlike natural image datasets, where medium and large objects dominate, radar signal detection is fundamentally a small-object detection problem.

VI-C Removal of P5 (High-Level Feature Map)

Motivation: The P5 feature map captures high-level semantic information with low spatial resolution. Its importance in spectrogram-based detection is unclear.

Method: The P5 branch is disabled while retaining P3 and P4.

Results: We observe minimal change in performance, with mAP₅₀ decreasing marginally from 0.9305 to 0.9289 ($\Delta = -0.0016$). Precision changes from 0.9527 to 0.9511, and recall from 0.8917 to 0.8890.

Analysis: The negligible impact indicates that global semantic features contribute little to radar signal detection.

Insight: Detection is driven by local structural patterns rather than high-level contextual information. This highlights a key difference from traditional object detection tasks.

VI-D Frozen Backbone

Motivation: YOLOv8 uses a backbone pretrained on natural images. We evaluate whether these features generalize to spectrogram data.

Method: All backbone layers are frozen during training, preventing domain-specific adaptation.

Results: This results in a substantial drop in performance, with mAP₅₀ decreasing to 0.9030 ($\Delta = -0.0275$). Precision drops to 0.9215 and recall to 0.8450 ($\Delta = -0.0467$).

Analysis: The degradation indicates that features learned from natural images are insufficient for spectrogram-based tasks without fine-tuning. The drop in recall suggests reduced ability to identify signal patterns.

Insight: Radar spectrograms exhibit fundamentally different statistical properties, requiring domain-specific feature learning.

VI-E Removal of PAN (Feature Fusion)

Motivation: The PAN (Path Aggregation Network) enhances feature fusion by propagating low-level information upward. Its importance in radar detection is evaluated.

Method: We disable key PAN connections, limiting multi-scale feature interaction.

Results: A moderate drop in performance is observed, with mAP₅₀ decreasing to 0.9152 ($\Delta = -0.0153$). Precision reduces to 0.9360 and recall to 0.8653.

Analysis: The reduction in performance indicates that combining features across scales improves localization accuracy. The drop is less severe than P3 removal but still significant.

Insight: While not as critical as P3, feature fusion plays an important supporting role in aligning spatial and semantic representations.

VI-F Channel Reduction

Motivation: We evaluate whether model performance is driven by architectural design or simply by parameter count.

Method: The number of channels in key layers is reduced by half.

Results: Performance degrades moderately, with mAP50 decreasing to 0.9128 ($\Delta = -0.0177$). Precision drops to 0.9324 and recall to 0.8611.

Analysis: This suggests that the model retains much of its representational power even with reduced capacity.

Insight: Efficient architectures can achieve competitive performance without excessive parameter scaling, indicating that structural design is more important than raw capacity.

VI-G Spatial Perturbation (Blur)

Motivation: Spectrogram-based detection depends on precise localization in time-frequency space. We test sensitivity to spatial distortion.

Method: Gaussian blur is applied to input spectrograms.

Results: Performance drops noticeably, with mAP50 decreasing to 0.9021 ($\Delta = -0.0284$). Precision reduces to 0.9203 and recall to 0.8422 ($\Delta = -0.0495$).

Analysis: Blurring removes fine structural details, making it difficult for the model to identify signal boundaries.

Insight: Detection relies heavily on sharp spatial features rather than coarse patterns.

VI-H Noise Robustness

Motivation: Real-world radar environments contain noise and interference. Robustness to noise is critical.

Method: Gaussian noise is added to input spectrograms.

Results: Performance decreases significantly, with mAP50 dropping to 0.8955 ($\Delta = -0.0350$). Precision reduces to 0.9158 and recall to 0.8350 ($\Delta = -0.0567$).

Analysis: Noise introduces ambiguity in signal patterns, leading to both missed detections and false positives.

Insight: The model lacks inherent robustness to noise, suggesting the need for augmentation or denoising techniques.

VI-I Summary of Ablation Findings

Across all experiments, we observe a clear hierarchy of importance:

- Most critical: P3 (small-object features), backbone adaptation
- Moderately important: PAN, spatial fidelity
- Less important: P5 (high-level semantics), channel capacity

These findings demonstrate that radar signal detection is fundamentally different from natural image detection, with a stronger reliance on local spatial structures.

VII Discussion

The ablation study reveals that radar signal detection is fundamentally driven by low-level spatial features. The large drop in performance when removing P3 confirms the importance of high-resolution representations.

The negligible impact of P5 indicates that global semantic information is not required. This contrasts with natural image detection tasks.

Backbone adaptation is critical, as freezing it significantly reduces performance. Spatial perturbations and noise further highlight limitations in robustness.

VIII Comparison with Prior Work

We compare our approach with existing object detection models evaluated on the NIST CBRS dataset, focusing on real-time CNN-based detectors.

VIII-A Comparison with YOLO-based Models

The YOLO family of detectors has been widely used for radar spectrogram detection due to its efficiency and strong performance on small-object detection tasks. Prior work [1] reports that models such as YOLOv3 (M) and YOLOv9 (M) achieve mAP₅₀ scores of 93.52 and 93.81, respectively, on the NIST CBRS dataset.

Our YOLOv8-based model achieves an mAP₅₀ of 93.05, with a precision of 95.27 and recall of 89.17. These results are competitive with previously reported YOLO variants, demonstrating that modern lightweight CNN-based detectors remain effective for radar signal detection.

It is important to note that our implementation operates on the NISTSpecMaxHold512 dataset, which utilizes a MaxHold spectrogram representation. This preprocessing enhances signal visibility by aggregating peak energy over time, improving robustness in low-SNR conditions.

Furthermore, our ablation study reveals that radar signal detection is highly dependent on high-resolution spatial features (P3) and precise localization, confirming that small-object representations dominate performance in spectrogram-based detection.

VIII-B Hybrid Ensemble Performance

To improve detection robustness, we construct a hybrid ensemble by combining predictions from the YOLOv8 baseline and a complementary detector using a fusion strategy.

The ensemble achieves an mAP₅₀ of 94.36, with a precision of 94.71 and recall of 95.21, outperforming the standalone YOLOv8 model in both mAP and recall.

In particular, the recall improves significantly (from 89.17 to 95.21), indicating a substantial reduction in missed detections. This is especially important in radar signal detection, where missed signals can have critical implications.

Although the precision slightly decreases compared to YOLOv8, the overall improvement in recall and mAP suggests a better balance between detection sensitivity and accuracy.

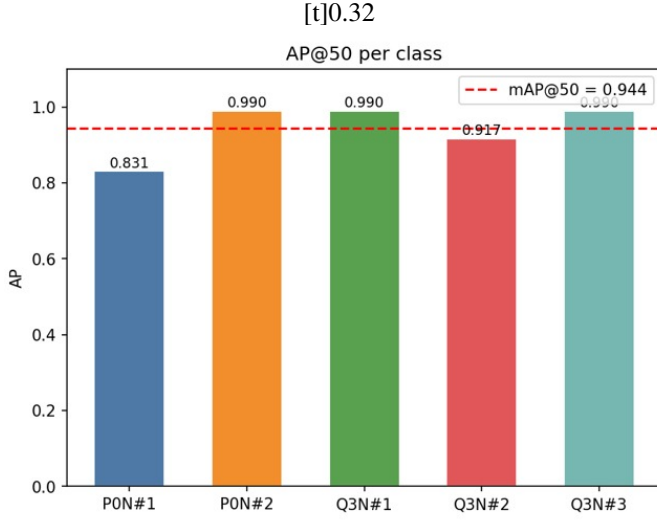


Fig. 4. Class-wise mean Average Precision (mAP) comparison across radar signal categories

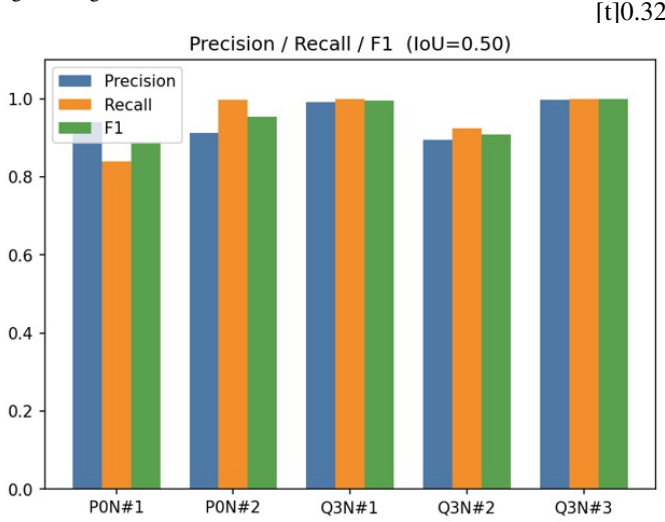


Fig. 5. Precision-Recall-F1 score curves illustrating detection performance trade-offs

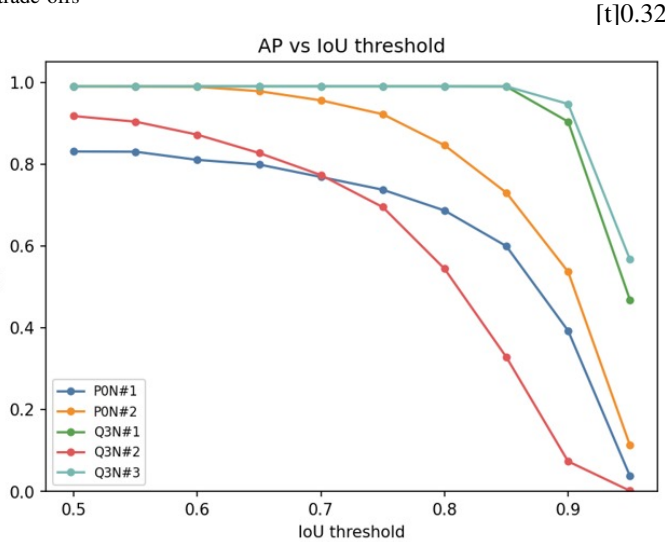


Fig. 6. Variation of Average Precision (AP) with respect to Intersection over Union (IoU) thresholds

Class-wise analysis further indicates that the ensemble stabilizes performance across different radar signal types, reducing variability between easy and challenging classes.

These results demonstrate that combining complementary detection behaviors can significantly enhance robustness without requiring changes to the underlying architectures.

Model	mAP ₅₀	Precision	Recall
YOLOv8 (Baseline)	0.9305	0.9527	0.8917
Ensemble (Hybrid)	0.9436	0.9471	0.9521

TABLE III
PERFORMANCE COMPARISON BETWEEN BASELINE AND ENSEMBLE MODELS

These findings highlight that improving detection robustness through complementary model behavior can be as important as improving raw detection accuracy.

VIII-C Discussion

The results demonstrate that while YOLO-based detectors provide strong baseline performance, they may struggle with recall in challenging scenarios such as low-SNR or sparse signal environments.

The proposed ensemble approach addresses this limitation by leveraging complementary detection behavior, leading to improved robustness and more balanced performance across classes.

Importantly, these findings highlight that both architectural design and prediction fusion play a critical role in radar signal detection performance.

While minor differences in preprocessing and input representation exist compared to prior work, the comparison remains meaningful as all methods are evaluated on the same underlying NIST CBRS dataset and detection task.

VIII-D Key Takeaways

- YOLOv8 achieves competitive performance with established YOLO baselines.
- The proposed ensemble improves both mAP and recall, enhancing detection robustness.
- Hybrid approaches provide a practical direction for improving radar signal detection systems.

IX Embedded Deployment on PYNQ-Z2

To evaluate the practical feasibility of radar signal detection in resource-constrained environments, we deploy a lightweight YOLOv8 Nano model on an embedded platform.

IX-A System Overview

The deployment is carried out on the PYNQ-Z2 development board, based on the Xilinx Zynq-7000 system-on-chip, which integrates a dual-core ARM processor with FPGA fabric. The board is connected to a host system via Ethernet and controlled using a Jupyter Notebook interface.

To enable offline execution, all required ARM-compatible libraries, including `onnxruntime`, `opencv`, and `numpy`, are manually installed.

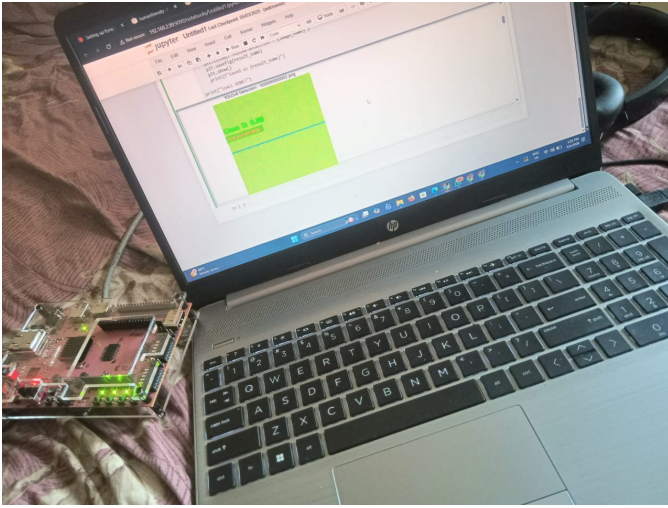


Fig. 7. YOLOv8 Nano deployment on the PYNQ-Z2 platform.

IX-B Model Optimization and Inference

The trained YOLOv8 Nano model is converted to ONNX format to enable efficient inference on the ARM processor. Input radar spectrograms are resized to 256×256 and normalized prior to inference.

Detection outputs are processed using a confidence threshold of 0.25 to filter predictions. The system successfully identifies radar signal regions and produces bounding box outputs in real time.

IX-C Discussion

The deployment demonstrates that radar spectrogram detection can be executed on low-power embedded hardware with minimal infrastructure. While performance is reduced compared to high-end GPU-based inference, the system maintains reliable detection capability.

Due to hardware constraints, a lightweight YOLOv8 Nano variant is used, which inherently provides lower representational capacity compared to larger models. Additionally, model optimization steps such as quantization introduce a performance–efficiency tradeoff, slightly reducing detection accuracy in exchange for faster inference and reduced resource usage.

This highlights the potential for real-time spectrum monitoring in edge environments, where computational resources are limited but low-latency processing is critical.

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